

ActInf Textbook Group ~ Cohort 5 ~ Session 8

(Chapter 4, Part 1) ~ 11/7/2023

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all right hello everyone cohort 5 it's our first discussion on chapter 4 so does anyone want to give just any thought on four overall just any any aspect of what they expected or got from it or however far they got or some section that was just interesting to them I have a question um so one kind of generative model that I am familiar with is um one way of representing a generative model is using a Markov decision process or partially observable Markov decision process uh this is for discrete um the discrete case of generative modeling I'm wondering what else what other what are other very popular or common uh structures that are used to uh write down or Express generative models anyone want to give a thought I would say one alternative way to write down parts of generative models is the continuous time but are you asking is there another way to represent discrete time generative models other than with a PDP right to begin with because of course there's the the whole other uh area of continuous time but um there's one important aspects to the whole um question of um of Bayesian inference which is or some some agent some organism that's implementing Bayesian inference is how does it uh not only learn the parameters of a generative model but how does it also learn the structure of that model um and this is connected for example to questions about nature versus nurture and and uh development in humans or animals like what are the um the initial conditions what kind of um either inductive priors or like um structure biases does our neural architecture uh provide us with from the very beginning and whatever it doesn't provide us with how how do we um figure out what kinds of structure is appropriate for modeling a particular domain all you have any thoughts uh well first of all uh about the initial uh states of any system uh that I mean it it doesn't always um in active inference modeling we don't we don't always assume U any agent like tabula rasa like I mean we don't always begin from scratch so uh in most cases we necessarily impose some initial conditions uh in order to make it more efficient to model the exact behavior of or phenomenon of interest uh because otherwise we won't be able to um to model the whole process I mean all the emergent properties that um the agent have

acquired the the agents have acquired through their lifespan uh so it is uh it wouldn't be feasible to try to um put everything in in such an encapsulated form of um generative model in order to Encompass uh the agents uh entire u i mean experience and lifespan so it is necessarily uh more focused than that uh and uh IT addresses a specific problem uh we're trying to uh we're trying to model so um I guess uh it's not uh a priori it's it's not something a priori uh and it isn't U any prerequisite to um model uh any situation but uh when it comes to how an agent acquires those uh initial learnings those initial States well obviously there are many many ways to uh address that uh problem U with with various perspectives but um of course F or active inference uh in my opinion is not um a theory or a framework that can answer every um every question regarding uh to the behavior of the agent uh in every possible situation so we need to look for uh other or other granular problems um in order to uh account for the lifespan experience of an agent got it uh thank you so um just to kind of reiterate and clarify this your response for myself um in my mind I kind of related a bit to on the one hand this distinction we we often need to make and keep in mind between an uh a realist interpretation of active inference and an instrumentalist one uh where the latter is us taking this this mindset of like okay uh there there are agents out there in the world there are things out there in the world and and they do stuff uh we can't be completely sure of exactly what they're implementing uh mechanistically but but we can develop a framework for uh making some SE sense of it for ourselves and part of the way that we make some progress on that uh making sense project is to use our own um suspicions use our own uh Paran knowledge to to write down or Express or postulate some some structure to postulates some generative model uh and then we can we can active reference as a template where we can input that generative model uh perform inference on it and then uh perhaps eventually make some predictions um about the behavior of a system and compare that to to what the system is actually doing but but what you're uh something that you're pointing out is that within the context of the framework uh there isn't necessarily some uh uh procedure for figuring out okay if you didn't have this uh these priors about what what it is that the generative model could look like uh and and I suspect ECT we suspect that uh for example humans don't come with all of the the structure that they need uh embedded from the moment they're born in order to model uh the world maybe they have some structure built in but they they acquire more of it and learn more of it over time um but active reference is is not doesn't have all the tools um to to perform that learning of the structure you that's also an area of of work that's you can do but it's not explicitly part of active ref uh just to clear up a little bit um here because you see even in the

the very fabric of the first principles of active inference namely the free energy principle so the free energy principle uh assumes the existence of quote unquote things and then try to characterize what would the existence of those things constitute okay so it it doesn't necessarily uh provide an account for uh I mean uh how the existence occurred how how I mean how how those emergent uh properties arise from um from from lower causalities of course it can describe uh those causal Relationships by assuming that those uh uh emergent phenomena uh have happened uh already so uh it takes as very first as the very first step and as an initial uh as an initial U State the very existence of those things right uh so when it comes to uh Pro U modeling um an agent's Behavior within active inference framework I believe the same mindset applies uh here as well so uh it doesn't necessarily account for the uh how questions but rather to um it it tries to characterize the what questions uh I'm not sure uh if it clears up the I mean your question here or not but um it's it's not an account for um it's not an account that uh tries to uh let me put it this way um if if we have this system at our disposal uh that we we try to understand um the its mechanism or or um the way it behaves active inference provides uh a comprehensive tool to uh write down uh the behavior of the agent uh within specific uh mathematical technology but it doesn't necessarily account for the um emergence of that agent it doesn't necessarily account for how that agent uh emerged at the first place uh so that's uh one of the misconceptions about free energy principle and uh active inference because some people uh think that uh free energy principle being the theory of every quote unquote thing would necessarily entail uh accounting for the emergence of uh those ontological entities uh but uh it it's it was never uh the claim of free energy principle so um yeah it takes as uh it takes the existence of those things um as the given and then try to describe those uh Behavior thank you um inks then Andrew hi I have a separate question but now I got involved in this so but I do want to understand so if we were to model like super deep thick temporal modeling of a person as they're born and they go through their life if someone was sitting there active inferencing everything that they're doing that would account for the development of the faster prior updating of priors and the slower updating of generative models right that would somehow build like a human with their understanding of the world not build but model how they become themselves okay well I mean to to the coarsest approximation yeah you could make a generative model that included decades would that mean it would give useful insights or interface with empirical data not not at that course graining could it be developed to be richer and more useful and meaningful interesting question but I don't know does anyone else want to give a thought on that well I would find it

useful I do have another question though I was just okay yeah okay sorry so this is um from the book page 77 um where the figure 44 describes the basian me basian message passing so my question there is the the black circle one the one that says intuitively current beliefs about States under each policy at each time are compared with those that would be predicted given beliefs about States at other times so this is the the statistician inferring by comparing different predicted States at different times does does that mean is basically time flowing left to right and you're trying all these different policy policies or are these alternative policies about something you can do right now and you can do different things about it and you're trying them out in your mind is this time flowing left to right or are these in my mind like if I do this this will happen if I do this this will happen and I compare them in my mind almost parall like they are counterfactuals to each other thank you yes Oli uh okay so there's a distinction and active inference literature between uh the time as uh t as the letter T and to so uh more often than not when active inference researchers use the letter t for time uh they uh they mean the flow of time as we know it I mean the intuitive uh time or our perceived time but uh what what but when they use T it doesn't necessarily refer to the flow of time but rather uh to uh to the steps uh that an agent uh would take in order to achieve a goal so uh to in a spec in a particular moment is not necessarily uh uh I mean the time interval between two tows are not necessarily equal right so uh it it tries to uh model the specific steps of uh the decision making or the action taken uh to achieve a particular goal thank you that's one aspect I think there's does anyone else want to give another thought on this and I clarify my question is are there counterfactual policies that only one will get implemented or is it step after step of of enacting like a project of long a strategy of many policies so in the pseudo code usually as implemented during the policy updating so going from the policy prior to the policy posterior that's using the expected free energy so that's when different policies are being considered so policy however long it is it's going to start with your next action and then include that many steps out so then if there's you know four options at each step four affordances so you're going to do policies of step two so there's 16 so then in the pseudo code it would have there's 16 policies and then there's the prior so if they're all equally likely then they be all even then all of them are iterated over in a loop and then while each one's getting iterated over it is rolled out and then then they're all compared at the end resulting in the updated distribution and then so let's just say it's it's um 80% for the first option and then 10 and then five then then a few so then you could draw from that distribution and choose to do that or you could pick just like the best option which is actually something that um Magnus discussed in this uh live stream 55.1

which is about this message passing because like there there it's kind of a lot to it but also it's a very important relationship and also one that's not really addressed in the textbook just a few times kind of dropped in uh okay Andrew and then anyone else yeah I just wanted to give my stab at uh the prior question um it seemed to be asking what active or the free energy principle is or isn't and um me it's a set of constraints it's it's telling you what open functions and Funk do namely your C's your Q's if I am correct um can and cannot do if they are to be an entity that persists spatial temporarily um so it it doesn't tell you what those functions are they it leaves them as three variables but it constrains their behavior so so it I guess it's a different way of thinking about things than um thinking about like what is force equals mass times acceleration it's not showing transfer of energy it's saying where energy is constrained to go if something is to persist over time yeah that's very interesting um also this this is one one paper the title kind of summarizes this very simple yet perplexing argument by axial constants it's not about what it takes it's about what took you there so it's pretty interesting um ra yeah can you elaborate what's uh your interpretation of that title is of what is of this title it's not about what it takes it's about what took you there I think here is said really well um so just to to generalize it um if something exists right now under the right conditions it's because it's done something like minimizing its free energy so it's as if it's used variational free energy however the question of whether it will exist tomorrow cannot be settled purely by resorting to the fact that it will minimize its free energy to get there so something could just all just you know it's I I don't it may also be like saying um like past results is not future performance or something like things could change in niche so it's it including it minimizing free energy right to the end so there's kind of a few other aspects but but um yeah and and related to as Oli mentioned earlier like conditioning on a thing existing by by choosing that kind of boundary of framework then there's questions that aren't being addressed and ones that are but yeah right so just to clarify my original um so definitely we in order to be able to apply any of this uh of these ideas we take for granted we take as uh as assumed the existence of mar conf BL kit which is what gives our definition of a thing um my question is about uh so already assuming that a thing exists and it has its its boundary uh internally it is representing is encoding some uh model of the generative process that's going on outside uh in the world and it um uses perception and action to um make sure that that model is as consistent as possible with its observations um in the world so my question is about what's inside that generative model um for example in the case of a pomdp which is one way of uh representing uh a discreet time generative model uh there's some structure to what that PP looks like the way that we write it

down and what what are the parameters um and I need to become more familiar with with the actual degrees of freedom so to speak the different things that you can uh add or or take away to construct a different varieties of of of a model but but let's just take that PDP um category as an example and the way that I am thinking about it right now what what you have is as you're um incorporating this this generative model you have some parameters and uh over time you update the parameters of through perception you update the parameters of that generative model to be more consistent uh to to reduce uh your free energy to minimize your free energy but my question is about the the the structure of that generative model not just the the parameters but for example if uh if it's a a basan network um that is describing the causal links between some uh hidden variables some causes leading causes in the world and like the observations that you're receiving that basan network has some structure there are different nodes and connections between that and uh somehow you have to figure out what should that uh Network look like uh should the connections between nodes be and what should each node represent um and and so I'm wondering if within the context of active inference there is uh there's a procedure there is procedure for exploring uh that structure or for given a thing that already exists how does it the its generative model evolve not only in terms of its parameters uh but more um uh generally in terms of its structure maybe this is related to the idea of hierarch models which uh I haven't uh learned much about yet but yeah maybe I hope that to Claris applies a bit more my question yeah two short answers first there's structure learning that is the whole question of modeling and that that's what the question of modeling is that's why it's just not done just by thinking about it because you have to fit the base graph to be capturing whatever factors you're modeling but also this paper and some other ones in that area approach the structure learning and also um in live stream 54 with the learning structure versus learning discussion that that um Toby uses excellent thank you did you link these somewhere um uh yeah just in in where the notes are the notes for chapter four four thank you so much yeah all right any other like thoughts on four or some section that some one found interesting if uh no one else has a question about four I have a question about uh inference algorithms but but I want to give everyone the chance to to ask your questions does anyone have like any overall thought on for yeah okay Andrew sure I I was still kind of reflecting on the the question earlier about message passing I feel like I'm going to mix some things up here but so that's back on figure 4.4 and I was just thinking about the question of counterfactuals um because I mean if um if an agent were to and you'd have to of course like model them you'd have to construct them in such a way that they were able to engage in planning which I

suppose would involve minimizing expected free energy G um so in that in that case whenever one is engaging in you know planning if we could model that um then I guess is someone I mean they'd be comparing counterfactuals in that case right they'd be comparing different potential policies that they have not already engaged in or or potentially are already underway um in some way but like you know maybe you're so far along the way and your series sequence of actions associated with a particular policy or you haven't started it yet and then from there the idea is you would go with the policy that has highest probability is it right you You' basically you'd go with the policy that minimizes expected free energy which would also like whenever we look at that in terms of a distribution over policies it' be the policy that has the highest probability which you end up necessarily going for um I'm just trying to because I'm slowly learning how important it is to specify when you're modeling like all these factors and things can get a little blurry whenever we start speaking a little bit more philosophically or from real life experience I'm just I don't know if all of that made sense right but U just throwing that out [Music] there just a quick uh clarification question uh you brought up contrafactual in relation to action planning and then uh Daniel wrote down here planning considering future times steps is all counterfactual uh what that brings to my mind is that the way that um we're thinking about inference sorry planning in active inference is uh Imagining the outcomes of different policies uh and rolling those out uh into the future and then Computing the expected free energy over those roll outs for each policy um and then you kind of score the different uh policies uh according to their expected free energy and this kind of gives you uh a distribution over which uh policy you're you you expect to take or which policy you believe you're you're going to take um but but it seems like you you maybe like you sample from that distribution so it's not like you you just uh take um the policy with with the the the lowest expected free energy if that's not true please someone dispel that yeah right yeah all planning is that's one kind of counterfactual counterfactual just being um considering something something that is not a factual like observable so anything about the future time steps is counterfactual um or like alternative pasts but then considering the future it's all counterfactuals and then this is also kind of from figure 5.4 um this is that section of um taking in habit and then either passing habit through the PO prior or updating it according to expected free energy and then but whether whether habit has passed through or there's an update um then the most likely path results or that distribution is sampled from yeah Jesse so these are two yeah these are two possible selection uh methods either maximum um posterior or or sampling those are two methods of policy selection given a policy posterior I guess for

me uh what I'm attempting to do is I mean because quite frankly it's all a bit over my head but um trying to pull this back from like a lot of this conversation you know we're we're trying to apply this to conscious processes which I think makes it a little bit more difficult in some respects because there's so much you know philosophical baggage um and so in the last section of this chapter even though I don't really understand the application of like the um no I'm sorry it wasn't the Lan uh approxim yeah the leace approximation uh on same motor reflexes and trying to think about this like even thinking about the mo the message framework uh in a purely physiological Paradigm where the message passing is say you know the sensory response from the you know the doctor hits your you know under your knee with a little rubber hammer and either you're going to have a reflex response or you're not right and so in that in that respect if I try to look at the the counterfactuals or the policies not as some you know conscious um process but as a purely phys excuse me my throat but is a purely a physiological one it just kind of I think for me makes it a little bit and maybe I'm on the wrong path and just Mis misunderstanding it but for me it's just making it a little bit uh more simplistic I guess to to deal with uh all the all the symbolisms and the idea of gradients and the idea of it's not like we're consciously always doing this and and you know undergoing the statistical process but this is you know essentially a way of understanding the probability that you know given certain factors and parameters you know your knee gets hit with a little rubber hammer and you or you don't so I guess I just want to kind of throw that out there for possible Direction yeah I mean it it's very true and relevant that that not not everything Maps or just people may have wildly different maps of what features or psychological experiences they map to like different intuitions about these Concepts so it's a huge space um Andrew then raw um I I don't want to be redundant uh Daniel pretty much summed up my point but it's yeah I think it's more about just thinking about it as far as like you're a like take the perspective of someone who's modeling um think of like you could produce a model of like a person or or some other phenomena whatever agent you're looking at and you could include planning as like an entire sort of Dimension so to speak or or layer to your model but that could just entirely be forone I imagine most people who are trying to model you know uh reflex arcs and looking at yeah the movement of joints in response to you know the pressure at this particular spot probably from an active inference approach I imagine would not include planning in the modeling process meanwhile of course if you're trying to model someone thinking and you know Consciousness or however close you want to come to that I think last week we discussed the the neurop phenomenology paper I could drop a link to that again but like that kind of

modeling certainly you could include planning as um as a dimension and and I bring this up again because it's just that's what I'm slowly learning with active inference is like while I enjoy the philosophical discussion I think it's super important to just think about like from a modeler's perspective even though that kind of like can remove some of the magic of of the process but it's um it's you choose what you're going to include or not include in the model I suppose right um I think I I just hope that that clarifies anything it was clarifying to me to start to understand that yeah just to connect to um Thomas pars guess stream um and like a little bit of the development of different generative models because chapter 4 is where the generative models being described seven and eight will go into more detail into the discreet and the continuous time but chapter four is kind of like just considering generative models um generally so the early implementations were in continuous time then people were thinking about how to get sequential Dynamics that's discussed in this textbook with these Predator pre type models so that is like where there was repetitive curse of writing so it was doing like repetitive structures but it was in a continuous State space of time um then they started going more into the explicit planning explicit like the pdps kind of tying back to the earlier discussions that we had so um and then later um The Continuous models were used at the more like sensory motor layer and then the discreet state space models at higher levels of decisionmaking just want to make that point Thomas explains more but yeah any any other like thoughts on four yeah I'd like to um chime in just a bit so I think it's worth noting that you this whole active inference is um is still very much in its infancy and you there's a lot to work out I mean you know when we talk about um and you to Jesse's point is that you there there is no Authority here I mean we're talking about how humans model their lives and so um the assumptions that not only the modeler has but also the person has is going to um you know bear on the counterfactuals and so you know to produce a model of a person you know we can't really discount the fact that what they're perceiving is not reality so I hear a lot of it assumptions that so I I question these assumptions and maybe it's because I'm coming out of worldly mapping but you there's just a lot of assumptions that you bring up questions for me um as to the accuracy which is yeah the intent of the whole project is to become more accurate thank you Susan one approach there because I I I really agree with with a lot of what you raised um by using similar terms and just being clear about what terms we're using then some Expressions can be evaluated and then and just sort of like very carefully understood and then different people may still make different claims about the same topics C certainly that's what discourse is right that's a very interesting aspect of this poort and The Institute more broadly

which is as I mentioned Susan this is very um stuff the art like a young young kind of project um and The Institute is part of the group of people who like everyone here in trying to understand the framework is participating in developing it and carrying it Forward uh but also something that's I know that the Institute is really um working on uh and keeping in mind is how to to optimize uh this process of of doing the science of thinking about the ideas uh collectively and as Daniel just mentioned part of that is uh to the extent that we uh engage with each other more often and have these discussions together uh and use um the most precise language that we can the ontology uh and become more familiar with the ontology then we start to become kind of like uh almost to approximation uh I have like a mathematical uh approach to thinking about it because in math you have very clear definitions about uh for everything that you're working with so so this is very helpful that um that we're we're becoming more familiar with the concepts and the terms and then using them um in a shared way in a way that hopefully makes sense to as many P people as possible in every session but it's also really excellent to have people who um are maybe less familiar with some of the technical details because uh then you can stop uh others and say like hey that's um isn't completely clear to me or I'm thinking of it from this more intuitive perspective can you um explain how my intuition laps into your to the terms that you're using yeah thank you um Andrew yeah quick follow on to that point um I was hoping the ontology on our um uh what do you call it our epistemological space had a lookup table for the um equation symbology and I didn't find that but it'd be really helpful if there was like a word or something I can do for lookup card to um help embed that ontology uh built into the active inference textbooks uh equation into my brain H it's a great idea it's definitely something we've considered um there's several steps that we know we'd have to do so like in our literature analysis from the end of 2022 we annotated the number of equations in each paper for for several hundred papers obviously we have that information for textbooks so determine where the equations are then just like we have for um some of the equations but not all of them um develop the natural language descriptions using the ontology terms that natural language description would be including all of the terms that you need to include make the mapping between the symbols and the ontology terms I mean it's a project that that ontology will include and other features pending slash people who want to do it thanks for pointing that out yeah but it would be epic and and if you want a sub page or if you want a form to do it in let's figure out how to do it yeah Susan um so if I'm premature in asking this question just you know just say so and we move on but before equations would become accurate wouldn't wouldn't it be um a matter of

um pattern languages would be more of a broad scope to be able to um you know audit the equations Spa that's a very interesting yeah lot lot of ways you could say it the equations are being evaluated according to kind of standard math so if we consider that as a pattern language like math is a pattern symbol set then and also the equations equations are usually trivially true because it's they're codifications of assumptions basically yeah they're manipulations of of what is brought in to the formalism right so then that has a certain epistemic status it's not nothing there are theoretic iCal and mathematical findings that that are they are real findings but that that doesn't justify or or imply direct application of to anything of of anything okay thank you yeah the on like making it more like a living textbook with rendering um equations with the natural language or change making the variables compatible or using the computer variables um those could help a lot otherwise there's a lot of variability in which notation are used and to what extent notation is described like hopefully to a limited extent in a textbook but still there's a lot of notation in the textbook and it it kind of like shifts among the CH the chapters just bringing in more variables not necessarily like using one core set of variables which would be hard to do but that's not what they do so one more question since we're at the top of the um would it be premature to um announce that we're we have a special guest that might challenge our assumptions on um physics Daniel I mean we'll we'll announce it when it's organized but yeah that sounds good anyone like have any thoughts on chapter 4 or like or what they want to ask about next week anyone have any other thoughts on four inks I just want to say I love this equ they're not equations but the ones that look like waterfalls that go in different patterns it's very inspiring to see them and just to Express gratitude for you guys you are Saints to do this with us to just sit here Daniel and try to figure out what the hell I mean when I say that and want to help us understand like I don't know that you guys get enough appreciation and gratitude so just saying thank you yeah it's awesome that people come out and and care and it's interesting topics and even without even understanding only at this level of zoomed out it's still is pretty cool I want to second that thank you Daniel yeah thank you 100 % it's amazing all you guys yeah everyone updating generative models learning about it we can only do like the tiniest amount in the synchronous times but you know yeah okay cool see you all next time for part two on chapter four thank you everyone great to see bye bye bye